

Financial Econometrics

Jean-Michel Zakoïan

ENSAE-CREST

CAPM: Models and tests

CAPM

- The CAPM (Capital Asset Pricing Model) was introduced by Sharpe (1964) [Nobel price in 1990] and Lintner (1965).
- Relies on the mean-variance portfolio theory of Markowitz (1959).
- Can be seen as the first step of the asset valuation theory.
- CAPM is an *equilibrium model*, i.e. based on the assumption that supply equals demand for the set of all assets.
- For the econometrician, the CAPM can be interpreted in terms of theoretical linear regressions of the net returns of individual assets on the net return of the "market portfolio".

- 1 Theoretical regression of the returns of baseline assets on the return of a portfolio
 - Regression on the return of an efficient portfolio
 - Systemic risk, diversification
- 2 CAPM
- 3 Estimation and tests of the CAPM

Setup

Let an economy with 1 riskless asset and N risky assets.

$\mathbf{z}_{t+1}$: vector of *net excess returns* of the N risky assets

$$\mathbf{z}_{t+1} = \mathbf{y}_{t+1} - (1 + r_t)\mathbf{e}, \quad \tilde{\boldsymbol{\mu}}_t = E_t(\mathbf{z}_{t+1}) = \boldsymbol{\mu}_t - (1 + r_t)\mathbf{e}$$

$z_{t+1}(\mathbf{a})$: *net excess return of the risky part* of a portfolio with composition $\mathbf{a}$ in risky assets

$$\begin{aligned} z_{t+1}(\mathbf{a}) &= \frac{\mathbf{a}'\mathbf{p}_{t+1}}{\mathbf{a}'\mathbf{p}_t} - (1 + r_t) \\ &= \frac{\mathbf{a}'\text{diag}(\mathbf{p}_t)\{\mathbf{z}_{t+1} + (1 + r_t)\mathbf{e}\}}{\mathbf{a}'\text{diag}(\mathbf{p}_t)\mathbf{e}} - (1 + r_t) \\ &= \frac{\mathbf{a}'\text{diag}(\mathbf{p}_t)\mathbf{z}_{t+1}}{\mathbf{a}'\text{diag}(\mathbf{p}_t)\mathbf{e}} = \frac{\boldsymbol{\omega}'_{\mathbf{a}}\mathbf{z}_{t+1}}{\boldsymbol{\omega}'_{\mathbf{a}}\mathbf{e}} := \bar{\boldsymbol{\omega}}'_{\mathbf{a}}\mathbf{z}_{t+1}, \end{aligned}$$

where $\boldsymbol{\omega}'_{\mathbf{a}} = (a_1 p_{1t}, \dots, a_N p_{Nt})$ and $\bar{\boldsymbol{\omega}}'_{\mathbf{a}} = \left(\frac{a_1 p_{1t}}{\sum_{j=1}^N a_j p_{jt}}, \dots, \frac{a_N p_{Nt}}{\sum_{j=1}^N a_j p_{jt}} \right)$

Theoretical linear regression

In particular, for an efficient portfolio we have

$$z_{t+1}(\mathbf{a}^*) = \frac{\tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \mathbf{z}_{t+1}}{\tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \mathbf{e}}.$$

Conditional on the available information at time t , the theoretical linear regression (with intercept) of the vector $\mathbf{z}_{t+1}$ on $z_{t+1}(\mathbf{a}^*)$ writes :

$$\mathbf{z}_{t+1} = \boldsymbol{\alpha}_t + \boldsymbol{\beta}_t z_{t+1}(\mathbf{a}^*) + \boldsymbol{\epsilon}_{t+1}$$

where

$$E_t(\boldsymbol{\epsilon}_{t+1}) = \mathbf{0}, \quad E_t(\boldsymbol{\epsilon}_{t+1} \boldsymbol{\epsilon}_{t+1}') = \boldsymbol{\Sigma}_t, \quad \text{Cov}_t(\boldsymbol{\epsilon}_{t+1}, z_{t+1}(\mathbf{a}^*)) = \mathbf{0},$$

$\boldsymbol{\alpha}_t, \boldsymbol{\beta}_t$ are $(N \times 1)$ vectors of parameters and $\boldsymbol{\Sigma}_t$ is a $N \times N$ matrix.

Nullity of the intercept in the regression

$z_{t+1}(\mathbf{a}^*)$: net excess return of an **efficient** portfolio.

Proposition

We have $\alpha_t = \mathbf{0}$. Hence

$$\mathbf{z}_{t+1} = \boldsymbol{\beta}_t z_{t+1}(\mathbf{a}^*) + \boldsymbol{\epsilon}_{t+1}.$$

*The components β_{tj} of vector $\boldsymbol{\beta}_t$ are called **betas** of the baseline assets on the efficient portfolio.*

Proof

$$z_{t+1}(\mathbf{a}^*) = \frac{\tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \mathbf{z}_{t+1}}{\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t} \Rightarrow \text{Var}_t(z_{t+1}(\mathbf{a}^*)) = \frac{\tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t}{[\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t]^2}.$$

From

$$\mathbf{z}_{t+1} = \boldsymbol{\alpha}_t + \boldsymbol{\beta}_t z_{t+1}(\mathbf{a}^*) + \boldsymbol{\epsilon}_{t+1}$$

we deduce

$$\begin{aligned} \boldsymbol{\beta}_t &= \text{Cov}_t(\mathbf{z}_{t+1}, z_{t+1}(\mathbf{a}^*)) [\text{Var}_t(z_{t+1}(\mathbf{a}^*))]^{-1} \\ &= \frac{\boldsymbol{\Omega}_t \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t [\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t]^2}{\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t \tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t} \\ &= \tilde{\boldsymbol{\mu}}_t \frac{\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t}{\tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t}, \end{aligned}$$

$$\boldsymbol{\alpha}_t = E_t(\mathbf{z}_{t+1}) - \boldsymbol{\beta}_t E_t\{z_{t+1}(\mathbf{a}^*)\} = \tilde{\boldsymbol{\mu}}_t - \boldsymbol{\beta}_t \frac{\tilde{\boldsymbol{\mu}}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t}{\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t} = 0.$$

□

A reciprocal

Proposition

Let a portfolio with composition $\mathbf{a}$ in risky assets.

In the regression

$$\mathbf{z}_{t+1} = \boldsymbol{\alpha}_t + \boldsymbol{\beta}_t \mathbf{z}_{t+1}(\mathbf{a}) + \boldsymbol{\epsilon}_{t+1},$$

the coefficient $\boldsymbol{\alpha}_t$ is null only if the portfolio is efficient.

Proof

We have $z_{t+1}(\mathbf{a}) = \frac{\omega'_a \mathbf{z}_{t+1}}{\omega'_a \mathbf{e}}$, thus

$$\boldsymbol{\beta}_t = \text{Cov}_t(\mathbf{z}_{t+1}, z_{t+1}(\mathbf{a})) [\text{Var}_t(z_{t+1}(\mathbf{a}))]^{-1} = \frac{\boldsymbol{\Omega}_t \omega_a \omega'_a \mathbf{e}}{\omega'_a \boldsymbol{\Omega}_t \omega_a},$$

$$\alpha_t = \tilde{\mu}_t - \boldsymbol{\beta}_t \frac{\tilde{\mu}'_t \omega_a}{\omega'_a \mathbf{e}} = \tilde{\mu}_t - \frac{\boldsymbol{\Omega}_t \omega_a \omega'_a \mathbf{e}}{\omega'_a \boldsymbol{\Omega}_t \omega_a} \frac{\tilde{\mu}'_t \omega_a}{\omega'_a \mathbf{e}}.$$

Hence $\alpha_t = 0$ iff

$$\tilde{\mu}_t \omega'_a \boldsymbol{\Omega}_t \omega_a = \boldsymbol{\Omega}_t \omega_a \tilde{\mu}'_t \omega_a.$$

Thus

$$\omega_a = \frac{1}{\lambda} \boldsymbol{\Omega}_t^{-1} \tilde{\mu}_t \quad \text{where} \quad \lambda = \frac{\omega'_a \boldsymbol{\Omega}_t \omega_a}{\omega'_a \tilde{\mu}_t}.$$

It follows that the portfolio is efficient.

□

Random walk for the log-prices

The coefficient β_t generally depends on the price at time t :

$$\beta_t = \tilde{\mu}_t \frac{\mathbf{e}' \boldsymbol{\Omega}_t^{-1} \tilde{\mu}_t}{\tilde{\mu}_t' \boldsymbol{\Omega}_t^{-1} \tilde{\mu}_t}.$$

An exception is given in the next result.

Corollary

If the log-prices follow a random walk,

$$\log p_{i,t+1} = \log p_{i,t} + u_{i,t+1}, \quad i = 1, \dots, N$$

where $(\mathbf{u}_t)$ is an iid sequence, with $\mathbf{u}_{t+1} = (u_{i,t+1})_{i=1,\dots,N}$ independent from $(\log p_{i,t})_{i=1,\dots,N}$, the vector β_t is non random.

If, in addition, $r_t = r, \forall t$, then

$$\beta_{tj} = \beta_j, \quad \forall t.$$

Proof

We have

$$p_{i,t+1} = p_{i,t} e^{u_{i,t+1}}.$$

Hence

$$E_t p_{i,t+1} = p_{i,t} E(e^{u_{i,t+1}}).$$

Hence $\boldsymbol{\mu}_t$ does not depend on $\mathbf{p}_t$ and $\tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t - (1 + r_t)\mathbf{e}$ is not random.

The same property holds for $\boldsymbol{\Omega}_t$.

If $r_t = r$, $\tilde{\boldsymbol{\mu}}_t$ and $\boldsymbol{\Omega}_t$ are constant.

□

Beta of a portfolio

The beta of a portfolio with composition $\mathbf{a}$ in risky assets is defined as

$$\beta_t(\mathbf{a}) = \frac{\omega'_a \boldsymbol{\beta}_t}{\omega'_a \mathbf{e}} := \bar{\omega}'_a \boldsymbol{\beta}_t.$$

The beta of a portfolio is thus an average of the baseline betas.

Decomposition of risk

For each risky asset i , we have

$$E_t(z_{i,t+1}) = \beta_{it} E_t\{z_{t+1}(\mathbf{a}^*)\}.$$

Moreover

$$\text{Var}_t(\mathbf{z}_{t+1}) = \boldsymbol{\beta}_t \boldsymbol{\beta}_t' \text{Var}_t\{z_{t+1}(\mathbf{a}^*)\} + \boldsymbol{\Sigma}_t.$$

Hence

$$\text{Var}_t(z_{i,t+1}) = \underbrace{\beta_{it}^2 \text{Var}_t\{z_{i,t+1}(\mathbf{a}^*)\}}_{\text{systemic risk}} + \underbrace{\sigma_{it}^2}_{\text{idiosyncratic risk}}.$$

Diversification reduces the idiosyncratic risk of a portfolio

Weight of the beta of the i th asset in the beta of the portfolio :

$$\bar{\omega}_{\mathbf{a},i} = \frac{a_i p_{i,t}}{\sum_{j=1}^N a_j p_{j,t}}$$

The global risk has the decomposition

$$\text{Var}_t(z_{t+1}(\mathbf{a})) = \beta_t^2(\mathbf{a}) \text{Var}_t\{z_{t+1}(\mathbf{a}^*)\} + \bar{\omega}'_{\mathbf{a}} \Sigma_t \bar{\omega}_{\mathbf{a}}.$$

Suppose for simplicity that $\Sigma_t = \sigma_t^2 I_N$, and that the portfolio has equal weights : $\bar{\omega}_{\mathbf{a},i} = 1/N$ for $i = 1, \dots, N$.

The idiosyncratic risk of this portfolio is thus

$$\bar{\omega}'_{\mathbf{a}} \Sigma_t \bar{\omega}_{\mathbf{a}} = \sum_{i=1}^N \bar{\omega}_{\mathbf{a},i}^2 \sigma_t^2 = \frac{\sigma_t^2}{N}.$$

When N increases, the importance of the idiosyncratic risk in the decomposition vanishes.

- 1 Theoretical regression of the returns of baseline assets on the return of a portfolio
- 2 **CAPM**
 - Equilibrium model
 - Regression on the market portfolio
 - Security Market Line
- 3 Estimation and tests of the CAPM

CAPM

The CAPM is a single-period *equilibrium model* between demand and supply in risky assets of a market.

Main result : Under the assumptions

- Investors have homogeneous anticipations and hold efficient portfolios (depending on their risk aversion)
- Absence of transaction costs
- Existence of a riskless asset

the portfolio built with the total wealth invested in risky assets - called **market portfolio**- is also **efficient**.

Equilibrium model

- $\mathbf{a}_t^{(m)}$: vector of units of risky assets available at time t
- The market portfolio has composition $\mathbf{a}_t^{(m)}$
- I investors

Each investor has a mean-variance optimization behavior. The risk aversion at time t of investor i is $\lambda_{i,t}$. His supply for risky assets is thus

$$\mathbf{a}_{i,t} = \frac{1}{\lambda_{i,t}} \{\text{diag}(\mathbf{p}_t)\}^{-1} \mathbf{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t.$$

The total supply for risky assets is then

$$\mathbf{a}_t = \sum_{i=1}^I \frac{1}{\lambda_{i,t}} \{\text{diag}(\mathbf{p}_t)\}^{-1} \mathbf{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t := \frac{1}{\bar{\lambda}_t} \{\text{diag}(\mathbf{p}_t)\}^{-1} \mathbf{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t$$

where

$$\bar{\lambda}_t = \left(\sum_{i=1}^I \frac{1}{\lambda_{i,t}} \right)^{-1}.$$

Equilibrium model

At the equilibrium, the available quantity of assets is equal to the total supply of the investors : thus

$$\mathbf{a}_t^{(m)} = \mathbf{a}_t = \frac{1}{\lambda_t} \{\text{diag}(\mathbf{p}_t)\}^{-1} \boldsymbol{\Omega}_t^{-1} \tilde{\boldsymbol{\mu}}_t.$$

It follows that **the market portfolio is efficient** and corresponds to a risk aversion $\bar{\lambda}_t$.

Remarks

At given available quantity $\mathbf{a}_t^{(m)}$ (which is exogenous in this model), the equilibrium is achieved by **price adjustment**.

The price adjustment mechanism is not explained by the model.

In practice, usual **proxies** of the market portfolio are the Standard and Poor S&P500 index or, on the French market, the CAC40 index.

Comments on the assumptions

The underlying assumptions can obviously be criticized. In particular :

- the possibility to make short operations, since no sign condition is imposed on the components of the vectors $\mathbf{a}_{i,t}$,
- the existence of a riskless asset,
- the fact that the investors perfectly know the conditional law of the prices (homogenous and rational anticipations),
- homogeneity of the investors.

Regression on the market portfolio

$z_{t+1}^{(m)}$: net excess return of the market portfolio.

Theoretical regression of $\mathbf{z}_{t+1}$ on $z_{t+1}^{(m)}$:

$$\mathbf{z}_{t+1} = \boldsymbol{\alpha}_t + \boldsymbol{\beta}_t z_{t+1}^{(m)} + \boldsymbol{\epsilon}_{t+1}$$

where

$$E_t(\boldsymbol{\epsilon}_{t+1}) = \mathbf{0}, \quad E_t(\boldsymbol{\epsilon}_{t+1} \boldsymbol{\epsilon}_{t+1}') = \boldsymbol{\Sigma}_t, \quad \text{Cov}_t(\boldsymbol{\epsilon}_{t+1}, z_{t+1}^{(m)}) = \mathbf{0}$$

and $\boldsymbol{\alpha}_t, \boldsymbol{\beta}_t$ are $(N \times 1)$ vectors of parameters and $\boldsymbol{\Sigma}_t$ is a $N \times N$ matrix.

Betas

A consequence of the CAPM is that $\alpha_t = 0$ in the regression :

$$\mathbf{z}_{t+1} = \boldsymbol{\beta}_t z_{t+1}^{(m)} + \boldsymbol{\epsilon}_{t+1}.$$

The components $\beta_{t,j}$ of the vector $\boldsymbol{\beta}_t$ are called **betas of the baseline assets on the market portfolio**.

The following relations between moments of the assets and the portfolio follow :

$$\tilde{\boldsymbol{\mu}}_t = \boldsymbol{\beta}_t E_t(z_{t+1}^{(m)}), \quad \boldsymbol{\Omega}_t = \boldsymbol{\beta}_t \text{Var}_t(z_{t+1}^{(m)}) \boldsymbol{\beta}_t' + \boldsymbol{\Sigma}_t,$$

$$\boldsymbol{\beta}_t = \frac{\text{Cov}_t(\mathbf{z}_{t+1}, z_{t+1}^{(m)})}{\text{Var}_t(z_{t+1}^{(m)})}.$$

Beta of a portfolio

The expected return of a portfolio **a** is

$$\tilde{\mu}_t(\mathbf{a}) := E_t(z_{t+1}(\mathbf{a})) = \beta_t(\mathbf{a})E_t(z_{t+1}^{(m)}) := \beta_t(\mathbf{a})\tilde{\mu}_t^{(m)},$$

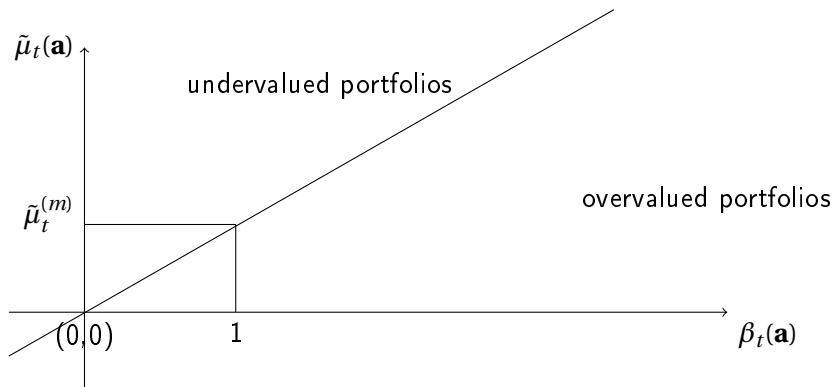
with $\beta_t(\mathbf{a}) = \overline{\omega}'_{\mathbf{a}}\boldsymbol{\beta}_t$. The risk premium of the portfolio is thus proportional to the expected return of the market portfolio.

$$\begin{aligned} z_{t+1}(\mathbf{a}) &= \beta_t(\mathbf{a})z_{t+1}^{(m)} + \overline{\omega}'_{\mathbf{a}}\boldsymbol{\epsilon}_{t+1}, \\ \text{Var}_t\{z_{t+1}(\mathbf{a})\} &= \beta_t^2(\mathbf{a})\text{Var}_t\{z_{t+1}^{(m)}\} + \overline{\omega}'_{\mathbf{a}}\boldsymbol{\Sigma}_t\overline{\omega}_{\mathbf{a}}. \end{aligned}$$

The proportionality coefficient - the beta of the portfolio - can be interpreted as both a **risk measure** and a **premium for holding risk**.

Security Market Line (SML)

Represented in the plan $(\beta_t(\mathbf{a}), \tilde{\mu}_t(\mathbf{a}))$ by the line $y = \tilde{\mu}_t^{(m)} x$.



Interpretation

A portfolio such that $\beta_t(\mathbf{a}) > 1$ can be called "**aggressive**" with respect to risk; conversely, a portfolio such that $\beta_t(\mathbf{a}) < 1$ can be called "**conservative**".

In the CAPM, **at the equilibrium**, portfolios lie on the market line.

Outside the equilibrium, portfolio can be over or under valued : they will be located below or above the market line.

Non invertibility of Σ_t

$$\mathbf{z}_{t+1} = \boldsymbol{\beta}_t \mathbf{z}_{t+1}^{(m)} + \boldsymbol{\epsilon}_{t+1}$$

is a **single-factor** model.

Because this factor, $\mathbf{z}_{t+1}^{(m)}$, is a linear combination of the baseline assets returns, Σ_t is not a full-rank matrix.

Proposition

Σ_t is singular, with $\text{rank}(\Sigma_t) = N - 1$ and

$$\Sigma_t = \mathbf{\Omega}_t - \frac{\mathbf{\Omega}_t \bar{\mathbf{w}}_m \bar{\mathbf{w}}_m' \mathbf{\Omega}_t}{\bar{\mathbf{w}}_m' \mathbf{\Omega}_t \bar{\mathbf{w}}_m},$$

where $\bar{\mathbf{w}}_m$ is the vector of allocations of the market portfolio (in value and %).

Proof (I)

We have $z_{t+1}^{(m)} = \bar{\omega}'_m \mathbf{z}_{t+1}$ and

$$\text{Var}_t(z_{t+1}^{(m)}) = \bar{\omega}'_m \mathbf{\Omega}_t \bar{\omega}_m, \quad \text{Cov}_t(\mathbf{z}_{t+1}, z_{t+1}^{(m)}) = \mathbf{\Omega}_t \bar{\omega}_m.$$

Moreover,

$$\mathbf{\Omega}_t = \boldsymbol{\beta}_t \text{Var}_t(z_{t+1}^{(m)}) \boldsymbol{\beta}'_t + \boldsymbol{\Sigma}_t, \quad \boldsymbol{\beta}_t = \frac{\text{Cov}_t(\mathbf{z}_{t+1}, z_{t+1}^{(m)})}{\text{Var}_t(z_{t+1}^{(m)})}.$$

Thus

$$\begin{aligned} \boldsymbol{\Sigma}_t &= \mathbf{\Omega}_t - \boldsymbol{\beta}_t \text{Var}_t(z_{t+1}^{(m)}) \boldsymbol{\beta}'_t \\ &= \mathbf{\Omega}_t - \frac{\mathbf{\Omega}_t \bar{\omega}_m}{\bar{\omega}'_m \mathbf{\Omega}_t \bar{\omega}_m} \bar{\omega}'_m \mathbf{\Omega}_t \bar{\omega}_m \frac{\bar{\omega}'_m \mathbf{\Omega}_t}{\bar{\omega}'_m \mathbf{\Omega}_t \bar{\omega}_m} = \mathbf{\Omega}_t - \frac{\mathbf{\Omega}_t \bar{\omega}_m \bar{\omega}'_m \mathbf{\Omega}_t}{\bar{\omega}'_m \mathbf{\Omega}_t \bar{\omega}_m} \end{aligned}$$

Proof (II)

Let $\mathbf{E} = \{\mathbf{x} \in \mathbb{R}^N | \mathbf{x}' \boldsymbol{\Sigma}_t \mathbf{x} = 0\}$. Then, $\text{rank}(\boldsymbol{\Sigma}_t) = N - \dim(\mathbf{E})$.

Then

$$\mathbf{x} \in \mathbf{E} \quad \Leftrightarrow \quad (\mathbf{x}' \boldsymbol{\Omega}_t \mathbf{x}) (\bar{\boldsymbol{\omega}}_m' \boldsymbol{\Omega}_t \bar{\boldsymbol{\omega}}_m) = \{\mathbf{x}' \boldsymbol{\Omega}_t \bar{\boldsymbol{\omega}}_m\}^2.$$

By Cauchy-Schwarz, with $\boldsymbol{\Omega}_t$ positive definite

$$\{\mathbf{x}' \boldsymbol{\Omega}_t \bar{\boldsymbol{\omega}}_m\}^2 \leq (\mathbf{x}' \boldsymbol{\Omega}_t \mathbf{x}) (\bar{\boldsymbol{\omega}}_m' \boldsymbol{\Omega}_t \bar{\boldsymbol{\omega}}_m)$$

with equality only if $\mathbf{x}$ and $\bar{\boldsymbol{\omega}}_m$ are colinear.

That is, $\dim(\mathbf{E}) = 1$.

Finally, $\boldsymbol{\Sigma}_t$ has rank $N - 1$.



β_t in function of the market portfolio

We obtained

$$\beta_t = \frac{\mathbf{\Omega}_t \bar{\mathbf{w}}_m}{\bar{\mathbf{w}}_m' \mathbf{\Omega}_t \bar{\mathbf{w}}_m}.$$

Assuming that $\mathbf{w}_{\mathbf{a}(m)}$ is observed, a natural estimator of this coefficient is thus

$$\hat{\beta} = \frac{\hat{\mathbf{\Omega}}_n \bar{\mathbf{w}}_m}{\bar{\mathbf{w}}_m' \hat{\mathbf{\Omega}}_n \bar{\mathbf{w}}_m},$$

where $\hat{\mathbf{\Omega}}_n$ is the empirical variance of the $\mathbf{y}_t$'s (also of the $\mathbf{z}_t$'s).

- 1 Theoretical regression of the returns of baseline assets on the return of a portfolio
- 2 CAPM
- 3 Estimation and tests of the CAPM
 - Estimation
 - Tests of the CAPM
 - Application

Regression on the market portfolio

Let

$$\mathbf{z}_{t+1} = (y_{1,t+1} - (1 + r_t), \dots, y_{M,t+1} - (1 + r_t))'$$

where $M \leq N$

(in practice, it is rare/impossible to include all the assets of a market).

Consider the regression of $\mathbf{z}_{t+1}$ on $\mathbf{z}_{t+1}(\mathbf{a})$:

$$\mathbf{z}_{t+1} = \boldsymbol{\alpha} + \boldsymbol{\beta} \mathbf{z}_{t+1}(\mathbf{a}) + \boldsymbol{\epsilon}_{t+1}, \quad t = 1, \dots, n$$

where

$$E_t(\boldsymbol{\epsilon}_{t+1} | \mathbf{z}_{t+1}(\mathbf{a})) = \mathbf{0}, \quad E_t(\boldsymbol{\epsilon}_{t+1} \boldsymbol{\epsilon}_{t+1}' | \mathbf{z}_{t+1}(\mathbf{a})) = \boldsymbol{\Sigma},$$

$\boldsymbol{\alpha}, \boldsymbol{\beta}$ are $(N \times 1)$ vectors of parameters and $\boldsymbol{\Sigma}$ is a $N \times N$ matrix.

We assume **serial uncorrelation** : $\text{Cov}(\boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_s) = \mathbf{0}$ for $t \neq s$.

SURE

We have

$$z_{i,t+1} = \alpha_i + \beta_i z_{t+1}(\mathbf{a}) + \epsilon_{i,t+1},$$
$$i = 1, \dots, m, \quad t = 1, \dots, n$$

→ System of regressions with the same explanatory variables 1 and $z_{t+1}(\mathbf{a})$ **for each asset**.

The system is referred to as *seemingly unrelated regressions* (SURE).

However, the error terms are correlated across assets :
can the equations be estimated asset by asset without efficiency loss ?

Rearranging the SURE equations

Let us arrange the equations **asset by asset** :

$$\mathbf{Z}_i = \begin{pmatrix} z_{i,1} \\ \vdots \\ z_{i,n} \end{pmatrix}, \quad \boldsymbol{\theta}_i = \begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix}, \quad \mathbf{e}_i = \begin{pmatrix} \epsilon_{i,1} \\ \vdots \\ \epsilon_{i,n} \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & \mathbf{z}_1(\mathbf{a}) \\ \vdots & \\ 1 & \mathbf{z}_n(\mathbf{a}) \end{pmatrix},$$

$$\Rightarrow \mathbf{Z}_i = \mathbf{X}\boldsymbol{\theta}_i + \mathbf{e}_i, \quad i = 1, \dots, m,$$

with

$$E(\mathbf{e}_i | \mathbf{X}) = 0, \quad \text{Cov}(\mathbf{e}_i, \mathbf{e}_j | \mathbf{X}) = \sigma_{ij} \mathbf{I}_n.$$

Rearranging the SURE equations

Then we stack the vectors $\mathbf{Z}_i$ as

$$\mathbf{Z} = \begin{pmatrix} \mathbf{Z}_1 \\ \vdots \\ \mathbf{Z}_m \end{pmatrix} = \begin{pmatrix} \mathbf{X} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{X} & \ddots & \mathbf{0} \\ \vdots & & \ddots & \vdots \\ \mathbf{0} & \cdots & \cdots & \mathbf{X} \end{pmatrix} \begin{pmatrix} \boldsymbol{\theta}_1 \\ \vdots \\ \vdots \\ \boldsymbol{\theta}_m \end{pmatrix} + \begin{pmatrix} \mathbf{e}_1 \\ \vdots \\ \vdots \\ \mathbf{e}_m \end{pmatrix} = (\mathbf{I}_m \otimes \mathbf{X})\boldsymbol{\theta} + \mathbf{e}.$$

The **Kronecker product** of 2 matrices is defined by

$$\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} a_{11}\mathbf{B} & \cdots & a_{1d_2}\mathbf{B} \\ \vdots & & \vdots \\ a_{d_1 1}\mathbf{B} & \cdots & a_{d_1 d_2}\mathbf{B} \end{pmatrix}.$$

We have $\mathbf{A} \otimes \mathbf{B} \neq \mathbf{B} \otimes \mathbf{A}$ in general,

$$(\mathbf{A} \otimes \mathbf{B})' = \mathbf{A}' \otimes \mathbf{B}', \quad (\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}, \quad (\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$$

if $\mathbf{A}$ and $\mathbf{B}$ are invertible

Rearranging the SURE equations

$$\mathbf{Z} = \begin{pmatrix} \mathbf{Z}_1 \\ \vdots \\ \mathbf{Z}_m \end{pmatrix} = \begin{pmatrix} \mathbf{X} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{X} & \ddots & \mathbf{0} \\ \vdots & & \ddots & \vdots \\ \mathbf{0} & \dots & \dots & \mathbf{X} \end{pmatrix} \begin{pmatrix} \boldsymbol{\theta}_1 \\ \vdots \\ \boldsymbol{\theta}_m \end{pmatrix} + \begin{pmatrix} \mathbf{e}_1 \\ \vdots \\ \mathbf{e}_m \end{pmatrix} = (\mathbf{I}_m \otimes \mathbf{X})\boldsymbol{\theta} + \mathbf{e}.$$

The GLS estimator of $\boldsymbol{\theta}$ is

► GLS

$$\hat{\boldsymbol{\theta}}_n = \{(\mathbf{I}_m \otimes \mathbf{X})' \boldsymbol{\Sigma}_\epsilon^{-1} (\mathbf{I}_m \otimes \mathbf{X})\}^{-1} (\mathbf{I}_m \otimes \mathbf{X})' \boldsymbol{\Sigma}_\epsilon^{-1} \mathbf{Z}$$

where

$$\boldsymbol{\Sigma}_\epsilon = V(\mathbf{e}) = \boldsymbol{\Sigma} \otimes \mathbf{I}_n.$$

GLS estimator

$$\begin{aligned}
 \hat{\theta}_n &= \{(\mathbf{I}_m \otimes \mathbf{X}')(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_n)(\mathbf{I}_m \otimes \mathbf{X})\}^{-1} (\mathbf{I}_m \otimes \mathbf{X}')(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_n)\mathbf{Z} \\
 &= (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{X}'\mathbf{X})^{-1} (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{X}')\mathbf{Z} \\
 &= (\boldsymbol{\Sigma} \otimes (\mathbf{X}'\mathbf{X})^{-1})(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{X}')\mathbf{Z} = (\mathbf{I}_m \otimes (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{Z} \\
 &= \begin{pmatrix} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' & 0 & \dots & 0 \\ 0 & (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' & \ddots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \dots & \dots & (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \end{pmatrix} \begin{pmatrix} \mathbf{Z}_1 \\ \vdots \\ \vdots \\ \mathbf{Z}_m \end{pmatrix}
 \end{aligned}$$

and we get the **LS estimators** for each asset

$$\hat{\theta}_{ni} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Z}_i, \quad i = 1, \dots, m.$$

Conclusion : estimating the system equation-by-equation **does not entail any efficiency loss**.

GLS estimators of α and β

$$\hat{\mu}_n = \frac{1}{n} \sum_{t=1}^n \mathbf{z}_t, \quad \hat{\mu}_n(\mathbf{a}) = \frac{1}{n} \sum_{t=1}^n z_t(\mathbf{a}), \quad \hat{\sigma}_n^2(\mathbf{a}) = \frac{1}{n} \sum_{t=1}^n (z_t(\mathbf{a}) - \hat{\mu}_n(\mathbf{a}))^2.$$

For the i th asset :

$$z_{i,t+1} = \alpha_i + \beta_i z_{t+1}(\mathbf{a}) + \epsilon_{i,t+1}, \quad t = 1, \dots, n$$

$$\begin{aligned} \Rightarrow \quad \hat{\alpha}_{i,n} &= \hat{\mu}_{i,n} - \hat{\beta}_{i,n} \hat{\mu}_n(\mathbf{a}), \\ \hat{\beta}_{i,n} &= \frac{\frac{1}{n} \sum_{t=1}^n (z_{i,t} - \hat{\mu}_{i,n})(z_t(\mathbf{a}) - \hat{\mu}_n(\mathbf{a}))}{\hat{\sigma}_n^2(\mathbf{a})}. \end{aligned}$$

Thus the GLS estimators of α and β :

$$\begin{aligned} \hat{\alpha}_n &= \hat{\mu}_n - \hat{\beta}_n \hat{\mu}_n(\mathbf{a}), \\ \hat{\beta}_n &= \frac{\frac{1}{n} \sum_{t=1}^n (\mathbf{z}_t - \hat{\mu}_n)(z_t(\mathbf{a}) - \hat{\mu}_n(\mathbf{a}))}{\hat{\sigma}_n^2(\mathbf{a})}. \end{aligned}$$

Estimation of Σ and MLE

$\Sigma = E(\epsilon_t \epsilon_t')$ can be estimated by

$$\hat{\Sigma}_n = \frac{1}{n} \sum_{t=1}^n \underbrace{\{\mathbf{z}_t - \hat{\alpha}_n - \hat{\beta}_n z_t(\mathbf{a})\}}_{\hat{\epsilon}_t} \underbrace{\{\mathbf{z}_t - \hat{\alpha}_n - \hat{\beta}_n z_t(\mathbf{a})\}'}_{\hat{\epsilon}_t'}.$$

Remark : If the distribution of $(\mathbf{z}'_{t+1}, z_{t+1}(\mathbf{a}))$ is Gaussian, hence $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \Sigma)$, with Σ positive definite, the log-likelihood writes

$$\begin{aligned} L(\alpha, \beta, \Sigma) &= -\frac{Mn}{2} \log(2\pi) - \frac{n}{2} \log|\Sigma| \\ &\quad - \frac{1}{2} \sum_{t=1}^n (\mathbf{z}_t - \alpha - \beta z_t(\mathbf{a}))' \Sigma^{-1} (\mathbf{z}_t - \alpha - \beta z_t(\mathbf{a})). \end{aligned}$$

The ML estimators of α and β coincide with the GLS estimators.

Asymptotic distribution of the GLS estimator

$$\mathbf{z}_{t+1} = \boldsymbol{\alpha} + \boldsymbol{\beta} z_{t+1}(\mathbf{a}) + \boldsymbol{\epsilon}_{t+1}, \quad t = 1, \dots, n$$

Assume that

- $\boldsymbol{\Omega}_t = \text{Var}_t(\mathbf{z}_{t+1}) = \boldsymbol{\Omega}$ (independent of t)
- $\mathbf{z}_{t+1} = \tilde{\boldsymbol{\mu}}_t + \boldsymbol{\Omega}^{1/2} \mathbf{U}_{t+1}$ where, $(\mathbf{U}_t) \stackrel{iid}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_M)$
- $\hat{\boldsymbol{\mu}}_n(\mathbf{a}) \rightarrow \tilde{\boldsymbol{\mu}}(\mathbf{a})$, $\hat{\sigma}_n^2(\mathbf{a}) \rightarrow \sigma^2(\mathbf{a})$ in probability as $n \rightarrow \infty$, for two constants $\tilde{\boldsymbol{\mu}}(\mathbf{a})$ and $\sigma^2(\mathbf{a}) > 0$.

Proposition

$$\sqrt{n} \begin{pmatrix} \hat{\boldsymbol{\alpha}}_n - \boldsymbol{\alpha} \\ \hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta} \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(0, \begin{pmatrix} \left(1 + \frac{\tilde{\boldsymbol{\mu}}^2(\mathbf{a})}{\sigma^2(\mathbf{a})} \right) \boldsymbol{\Sigma} & \frac{-\tilde{\boldsymbol{\mu}}(\mathbf{a})}{\sigma^2(\mathbf{a})} \boldsymbol{\Sigma} \\ \frac{-\tilde{\boldsymbol{\mu}}(\mathbf{a})}{\sigma^2(\mathbf{a})} \boldsymbol{\Sigma} & \frac{1}{\sigma^2(\mathbf{a})} \boldsymbol{\Sigma} \end{pmatrix} \right).$$

Recall that $\frac{\tilde{\boldsymbol{\mu}}^2(\mathbf{a})}{\sigma^2(\mathbf{a})} = S_t^2(\mathbf{a})$.

Sketch of the proof :

$$\mathbf{z}_t = \boldsymbol{\alpha} + \boldsymbol{\beta} z_t(\mathbf{a}) + \boldsymbol{\epsilon}_t \quad \Rightarrow \quad \hat{\boldsymbol{\mu}}_n = \boldsymbol{\alpha} + \boldsymbol{\beta} \hat{\mu}_n(\mathbf{a}) + \bar{\boldsymbol{\epsilon}}_n$$

$$\hat{\boldsymbol{\beta}}_n = \frac{\frac{1}{n} \sum_{t=1}^n (\mathbf{z}_t - \hat{\boldsymbol{\mu}}_n)(z_t(\mathbf{a}) - \hat{\mu}_n(\mathbf{a}))}{\hat{\sigma}_n^2(\mathbf{a})}, \quad \hat{\sigma}_n^2(\mathbf{a}) = \frac{1}{n} \sum_{t=1}^n (z_t(\mathbf{a}) - \hat{\mu}_n(\mathbf{a}))^2$$

We thus have

$$\begin{aligned} \sqrt{n}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}) &= \sqrt{n} \frac{\frac{1}{n} \sum_{t=1}^n \{\mathbf{z}_t - \hat{\boldsymbol{\mu}}_n - \boldsymbol{\beta}(z_t(\mathbf{a}) - \hat{\mu}_n(\mathbf{a}))\}(z_t(\mathbf{a}) - \hat{\mu}_n(\mathbf{a}))}{\hat{\sigma}_n^2(\mathbf{a})} \\ &= \sqrt{n} \frac{\frac{1}{n} \sum_{t=1}^n \{\boldsymbol{\epsilon}_t - \bar{\boldsymbol{\epsilon}}_n\}(z_t(\mathbf{a}) - \hat{\mu}_n(\mathbf{a}))}{\hat{\sigma}_n^2(\mathbf{a})} \\ &\stackrel{d}{=} \frac{1}{\sigma^2(\mathbf{a})} \frac{1}{\sqrt{n}} \sum_{t=1}^n \boldsymbol{\epsilon}_t \tilde{z}_t(\mathbf{a}), \end{aligned}$$

where $\tilde{z}_t(\mathbf{a}) = z_t(\mathbf{a}) - \tilde{\mu}(\mathbf{a})$ (using also : $\hat{\mu}_n(\mathbf{a}) \rightarrow \tilde{\mu}(\mathbf{a})$ and $\bar{\boldsymbol{\epsilon}}_n \rightarrow 0$)

Sketch of the proof :

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}) \stackrel{d}{=} \frac{1}{\sigma^2(\mathbf{a})} \frac{1}{\sqrt{n}} \sum_{t=1}^n \boldsymbol{\epsilon}_t \tilde{z}_t(\mathbf{a}),$$

Moreover, since $\hat{\mu}_n = \boldsymbol{\alpha} + \boldsymbol{\beta} \hat{\mu}_n(\mathbf{a}) + \bar{\boldsymbol{\epsilon}}_n = \hat{\boldsymbol{\alpha}}_n + \hat{\boldsymbol{\beta}}_n \hat{\mu}_n(\mathbf{a}),$

$$\sqrt{n}(\hat{\boldsymbol{\alpha}}_n - \boldsymbol{\alpha}) \stackrel{d}{=} \sqrt{n} \bar{\boldsymbol{\epsilon}}_n - \sqrt{n} \hat{\mu}_n(\mathbf{a}) (\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta})$$

Thus

$$\sqrt{n} \begin{pmatrix} \hat{\boldsymbol{\alpha}}_n - \boldsymbol{\alpha} \\ \hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta} \end{pmatrix} = \begin{pmatrix} I_M & \frac{-\tilde{\mu}(\mathbf{a})}{\sigma^2(\mathbf{a})} I_M \\ 0 & \frac{1}{\sigma^2(\mathbf{a})} I_M \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{n}} \sum_{t=1}^n \boldsymbol{\epsilon}_t \\ \frac{1}{\sqrt{n}} \sum_{t=1}^n \boldsymbol{\epsilon}_t \tilde{z}_t(\mathbf{a}) \end{pmatrix} := \mathbf{A} \mathbf{X}_n.$$

Moreover, it can be shown that

$$\mathbf{X}_n \xrightarrow{d} \mathcal{N} \left(0, \begin{pmatrix} \boldsymbol{\Sigma} & 0 \\ 0 & \sigma^2(\mathbf{a}) \boldsymbol{\Sigma} \end{pmatrix} \right).$$

Gaussian case

When $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \Sigma)$, it can be shown that

- i) $\hat{\alpha}_n$ and $\hat{\Sigma}_n$ are independent;
- ii) The distribution at fixed n of $n\hat{\Sigma}_n$ is a **Wishart** $W(\Sigma, n-2)$.

What is a Wishart distribution?

If $\mathbf{Z}_1, \dots, \mathbf{Z}_n$ are $p \times 1$ iid vectors which are $\mathcal{N}(\mathbf{0}, \Sigma)$ distributed, the random $p \times p$ matrix $\mathbf{A} = \sum_{i=1}^n \mathbf{Z}_i \mathbf{Z}_i'$ follows a Wishart law $W(\Sigma, n)$.

For $p=1$ and $\Sigma=1$ we retrieve the $\chi^2(n)$ distribution.

CAPM regression

We now consider the regression of $\mathbf{z}_{t+1}$ (vector of the net returns of M assets, with $M \leq N$) on $z_{t+1}^{(m)}$,

$$\mathbf{z}_{t+1} = \boldsymbol{\alpha} + \boldsymbol{\beta} z_{t+1}^{(m)} + \boldsymbol{\epsilon}_{t+1}, \quad t = 1, \dots, n$$

where

$$E_t(\boldsymbol{\epsilon}_{t+1}) = \mathbf{0}, \quad E_t(\boldsymbol{\epsilon}_{t+1} \boldsymbol{\epsilon}_{t+1}') = \boldsymbol{\Sigma}, \quad \text{Cov}(\boldsymbol{\epsilon}_{t+1}, z_{t+1}^{(m)}) = \mathbf{0}$$

and $\boldsymbol{\alpha}, \boldsymbol{\beta}$ are $(M \times 1)$ vectors of parameters and $\boldsymbol{\Sigma}$ a $M \times M$ matrix.

Hypotheses and interpretation

We wish to test a consequence of the CAPM, that is

$$H_0 : \alpha = 0,$$

against the alternative hypothesis

$$H_1 : \alpha \neq 0.$$

Remark : A test of CAPM can receive two interpretations :

- If we assume that the asset used as the market portfolio (for instance the CAC index) is efficient, we wish to test if the assets are correctly valuated.
- Conversely, if we assume that the price of the baseline assets are correctly fixed by the demand/supply mechanism, what is tested is the efficiency of the "market portfolio".

Wald test

The Wald test uses a consistent estimator of the asymptotic variance-covariance matrix of $(\hat{\alpha}_n, \hat{\beta}_n)$, given by

$$\begin{pmatrix} \left(1 + \frac{\hat{\mu}_n^{(m)2}}{\hat{\sigma}_n^{(m)2}}\right) \hat{\Sigma}_n & \frac{-\hat{\mu}_n^{(m)}}{\hat{\sigma}_n^{(m)2}} \hat{\Sigma}_n \\ \frac{-\hat{\mu}_n^{(m)}}{\hat{\sigma}_n^{(m)2}} \hat{\Sigma}_n & \frac{1}{\hat{\sigma}_n^{(m)2}} \hat{\Sigma}_n \end{pmatrix}$$

where

$$\hat{\mu}_n^{(m)} = \frac{1}{n} \sum_{t=1}^n z_t^{(m)}, \quad \hat{\sigma}_n^{(m)2} = \frac{1}{n} \sum_{t=1}^n \{z_t^{(m)} - \hat{\mu}_n^{(m)}\}^2$$

$$\hat{\Sigma}_n = \frac{1}{n} \sum_{t=1}^n \{z_t - \hat{\alpha}_n - \hat{\beta}_n z_t^{(m)}\} \{z_t - \hat{\alpha}_n - \hat{\beta}_n z_t^{(m)}\}'.$$

It is important to distinguish cases where Σ is invertible (which entails invertibility of $\hat{\Sigma}_n$ for n large enough) from cases where it is singular (in particular when $M = N$).

Wald test when Σ is nonsingular

The Wald statistic for H_0 is

$$W_n = \left(1 + \frac{\hat{\mu}_n^{(m)2}}{\hat{\sigma}_n^{(m)2}} \right)^{-1} n \hat{\alpha}'_n \hat{\Sigma}_n^{-1} \hat{\alpha}_n.$$

Under H_0 ,

$$W_n \xrightarrow{d} \chi_M^2.$$

The **critical region**, at the asymptotic level $\alpha \in (0, 1)$, is thus

$$C = \{W_n > \chi_M^2(1 - \alpha)\}$$

where $\chi_M^2(1 - \alpha)$ the $(1 - \alpha)$ quantile of the χ_M^2 law.

Remarks

- The term $\frac{\hat{\mu}_n^{(m)2}}{\hat{\sigma}_n^{(m)2}}$ is the square of the estimated Sharpe ratio of the market portfolio.
- The test is **consistent**.

Under H_1 : $\alpha \neq 0$,

$$W_n = \underbrace{n \left(1 + \frac{\hat{\mu}_n^{(m)2}}{\hat{\sigma}_n^{(m)2}} \right)^{-1} \hat{\alpha}'_n \hat{\Sigma}_n^{-1} \hat{\alpha}_n}_{\xrightarrow{n \rightarrow \infty} w = \left(1 + \frac{\bar{\mu}^{(m)2}}{\bar{\sigma}^{(m)2}} \right)^{-1} \alpha' \Sigma^{-1} \alpha > 0} \xrightarrow{n \rightarrow \infty} \infty.$$

Thus

$$P_{H_1}(C) \rightarrow 1, \quad \text{as} \quad n \rightarrow \infty.$$

Remarks

- The **power of the test** can be studied.

In the Gaussian case

$$\frac{n-M-1}{M(n-2)} W_n \sim \text{noncentral F-distribution}^*$$

with degrees of freedom M and $n-M-1$. The noncentrality parameter is

$$w_n = \left(1 + \frac{\mu^{(m)2}}{\sigma^{(m)2}} \right)^{-1} n \alpha \Sigma^{-1} \alpha.$$

*. If X and Y are independent, where X follows a noncentral chi-squared distribution with n_1 degrees of freedom and noncentrality parameter λ , and $Y \sim \chi^2(n_2)$, the ratio $\frac{X/n_1}{Y/n_2}$ follows a noncentral F-distribution.

Testing the nullity of one coefficient

To test the nullity of one coefficient α_i , a Student test can be used.

The critical region at the asymptotic level α is

$$\{|t_n| > \Phi^{-1}(1 - \alpha/2)\}$$

where

$$t_n = \left(1 + \frac{\hat{\mu}_n^{(m)2}}{\hat{\sigma}_n^{(m)2}}\right)^{-1/2} \frac{\sqrt{n}\hat{\alpha}_{ni}}{\hat{\sigma}_{ni}}$$

and $\hat{\sigma}_{ni}^2$ is the i th diagonal term of $\hat{\Sigma}_n$.

Wald test when Σ is singular

When $N = M$, $\text{rank}(\Sigma) = N - 1$.

The statistic W_n should not be used because $\hat{\Sigma}_n$ is close to be nonsingular for n large.

In this case, $\sqrt{n}(\hat{\alpha}_n - \alpha)$ asymptotically belongs to an hyperplan. It is thus possible to find a linear transformation of this vector which has an asymptotic non degenerate Gaussian distribution.

A classical method consists in using a [generalized inverse](#) of Σ .

g-inverses

If $\mathbf{A}$ is a $m \times n$ matrix, $\mathbf{A}^-$ is a g-inverse of $\mathbf{A}$ if $\mathbf{AA}^-\mathbf{A} = \mathbf{A}$.

A g-inverse always exists but may be not unique.

If $\mathbf{A}'\mathbf{A}$ is invertible, a g-inverse of $\mathbf{A}$ is $(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'$.

If $\mathbf{A}$ is diagonal,

$$\mathbf{A} = \begin{pmatrix} \mathbf{\Lambda} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}, \quad \mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \ddots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \dots & \dots & \lambda_m \end{pmatrix}, \quad \lambda_i \neq 0,$$

then

$$\mathbf{A}^- = \begin{pmatrix} \mathbf{\Lambda}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{D} \end{pmatrix} \quad \text{where } \mathbf{D} \text{ is any diagonal matrix.}$$

g-inverses

If $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$, where $\mathbf{\Sigma}$ is a $N \times N$ matrix with $\text{rk}(\mathbf{\Sigma}) = N_1 < N$.

We have $\mathbf{\Sigma} = \mathbf{P}' \mathbf{D} \mathbf{P}$ where $\mathbf{P}$ is orthogonal and $\mathbf{D} = \text{diag}(d_1, \dots, d_{N_1}, 0, \dots, 0)$ is diagonal.

Then a g-inverse of $\mathbf{\Sigma}$ is

$$\mathbf{\Sigma}^- = \mathbf{P}' \mathbf{D}^- \mathbf{P}.$$

For $\mathbf{D}^- = \text{diag}(d_1^{-1}, \dots, d_{N_1}^{-1}, 0, \dots, 0)$,

$$\mathbf{X}' \mathbf{\Sigma}^- \mathbf{X} = \mathbf{X}' \mathbf{P}' (\mathbf{D}^-)^{1/2} (\mathbf{D}^-)^{1/2} \mathbf{P} \mathbf{X}$$

and

$$\mathbf{Y} := (\mathbf{D}^-)^{1/2} \mathbf{P} \mathbf{X} \sim \mathcal{N}(\mathbf{0}, \underbrace{(\mathbf{D}^-)^{1/2} \mathbf{D} (\mathbf{D}^-)^{1/2}}_{\text{diag}(1, \dots, 1, 0, \dots, 0)})$$

Thus $\mathbf{Y} \stackrel{d}{=} (Y_1, \dots, Y_{N_1}, 0, \dots, 0)'$ where $Y_i \sim \mathcal{N}(0, 1)$. Finally

$$\mathbf{Y}' \mathbf{Y} = \mathbf{X}' \mathbf{\Sigma}^- \mathbf{X} \sim \chi^2(N_1).$$

Wald test when Σ is singular

A Wald statistic is

$$\tilde{W}_n = \left(1 + \frac{\hat{\mu}_n^{(m)2}}{\hat{\sigma}_n^{(m)2}} \right)^{-1} n \hat{\alpha}'_n \hat{\Sigma}_n^* \hat{\alpha}_n.$$

where $\hat{\Sigma}_n^*$ is a **consistent estimator of a g-inverse** Σ^- of Σ .

It can be shown that, under H_0 ,

$$\tilde{W}_n \xrightarrow{d} \chi^2(N-1).$$

Wald test when $N = M$

In particular, if

$$\Sigma = \Omega - \frac{\Omega \omega_{\mathbf{a}^{(m)}} \omega'_{\mathbf{a}^{(m)}} \Omega}{\omega'_{\mathbf{a}^{(m)}} \Omega \omega_{\mathbf{a}^{(m)}}},$$

a generalized inverse of Σ is Ω^{-1} :

$$\begin{aligned} \Sigma \Omega^{-1} \Sigma &= \left(\Omega - \frac{\Omega \omega_{\mathbf{a}^{(m)}} \omega'_{\mathbf{a}^{(m)}} \Omega}{\omega'_{\mathbf{a}^{(m)}} \Omega \omega_{\mathbf{a}^{(m)}}} \right) \Omega^{-1} \left(\Omega - \frac{\Omega \omega_{\mathbf{a}^{(m)}} \omega'_{\mathbf{a}^{(m)}} \Omega}{\omega'_{\mathbf{a}^{(m)}} \Omega \omega_{\mathbf{a}^{(m)}}} \right) \\ &= \Omega - 2 \frac{\Omega \omega_{\mathbf{a}^{(m)}} \omega'_{\mathbf{a}^{(m)}} \Omega}{\omega'_{\mathbf{a}^{(m)}} \Omega \omega_{\mathbf{a}^{(m)}}} + \frac{\Omega \omega_{\mathbf{a}^{(m)}} (\omega'_{\mathbf{a}^{(m)}} \Omega \Omega^{-1} \Omega \omega_{\mathbf{a}^{(m)}}) \omega'_{\mathbf{a}^{(m)}} \Omega}{(\omega'_{\mathbf{a}^{(m)}} \Omega \omega_{\mathbf{a}^{(m)}})^2} \\ &= \Omega - \frac{\Omega \omega_{\mathbf{a}^{(m)}} \omega'_{\mathbf{a}^{(m)}} \Omega}{\omega'_{\mathbf{a}^{(m)}} \Omega \omega_{\mathbf{a}^{(m)}}} \\ &= \Sigma \end{aligned}$$

Wald test when $N = M$

In particular, if

$$\Sigma = \Omega - \frac{\Omega \omega_{\mathbf{a}^{(m)}} \omega_{\mathbf{a}^{(m)}}' \Omega}{\omega_{\mathbf{a}^{(m)}}' \Omega \omega_{\mathbf{a}^{(m)}}},$$

a generalized inverse of Σ is Ω^{-1} .

A Wald statistic for testing H_0 when the returns of all the assets are regressed on the market portfolio is thus

$$\tilde{W}_n = \left(1 + \frac{\hat{\mu}_n^{(m)2}}{\hat{\sigma}_n^{(m)2}} \right)^{-1} n \hat{\alpha}_n' \hat{\Omega}_n^{-1} \hat{\alpha}_n.$$

Under H_0 , the statistic $\tilde{W}_n$ follows asymptotically a $\chi^2(N-1)$.

Inconsistency of the generalized inverse of an estimator

⚠ Using the generalized inverse $\hat{\Sigma}_n^-$ of a consistent estimator $\hat{\Sigma}_n$ of a variance matrix Σ **may yield an inconsistent estimator** of Σ^- !

Generally we do not have $\hat{\Sigma}_n^- \rightarrow \Sigma^-$.

Example :

$$\Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \hat{\Sigma}_n = \begin{pmatrix} 1 & \frac{1}{n} \\ \frac{1}{n} & \frac{\tau}{n^2} \end{pmatrix}, \quad \text{where } \tau \neq 1.$$

$\hat{\Sigma}_n^{-1} = \frac{1}{\tau-1} \begin{pmatrix} \tau & -n \\ -n & n^2 \end{pmatrix}$ **does not converge** to any g-inverse of Σ .

Moreover, if $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \Sigma)$, we have $\mathbf{X}' \hat{\Sigma}_n^{-1} \mathbf{X} = \frac{\tau}{\tau-1} X_1^2$ which does not follow a $\chi^2(1)$.

2-inverses

When a **consistent** estimator of Σ^- is **not available**, an alternative consists in using **{2}-inverses** instead of generalized inverses.

If A is a $m \times n$ matrix, A_- is a 2-inverse of A if $A_- A A_- = A_-$.

For $\Sigma = P D P'$ where $D = \text{diag}(\lambda_1, \dots, \lambda_N)$, with $\lambda_1 \geq \dots \geq \lambda_N$, let

$$D_-^{(k)} = \text{diag}(\lambda_1^{-1}, \dots, \lambda_k^{-1}, 0, \dots, 0), \quad k \leq \text{rank}(\Sigma).$$

Then $\Sigma_-^{(k)} = P D_-^{(k)} P'$ is a **2-inverse** of Σ :

$$\bullet \quad \Sigma_-^{(k)} \Sigma \Sigma_-^{(k)} = P D_-^{(k)} P' P D P' P' D_-^{(k)} P = P \underbrace{D D_-^{(k)} D}_{D_-^{(k)}} P \Sigma_-^{(k)}$$

but **not a g-inverse** of Σ :

$$\bullet \quad \Sigma \Sigma_-^{(k)} \Sigma = P D P' P D_-^{(k)} P' P D P' = P \underbrace{D D_-^{(k)} D}_{\text{diag}(\lambda_1, \dots, \lambda_k, 0, \dots, 0)} P' \neq \Sigma$$

2-inverses

It can be shown that

if $\hat{\Sigma}_n \rightarrow \Sigma$, then $\hat{\Sigma}_{n-}^{(k)} \rightarrow \Sigma_-^{(k)}$ if $k \leq \text{rank}(\Sigma)$.

Thus,

$$\text{if } \mathbf{X} \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad \mathbf{X}' \hat{\Sigma}_{n-}^{(k)} \mathbf{X} \xrightarrow{d} \chi^2(k).$$

In practice, the eigenvalues of $\hat{\Sigma}_n$ that are sufficiently far from 0 are inverted.

 If k is chosen too large the constraint $k \leq \text{rank}(\Sigma)$ might not be satisfied; if it is too small, the test will have low power.

Likelihood Ratio (LR) test

The LR test aims at assessing if the constraint (i.e., the null hypothesis) is supported by the data, by comparing the constrained and unconstrained likelihoods of the model

$$\mathbf{z}_t = \boldsymbol{\alpha} + \boldsymbol{\beta} z_t^{(m)} + \boldsymbol{\epsilon}_t, \quad t = 1, \dots, n$$

In this part we assume that $\boldsymbol{\Sigma} = \text{Var}(\boldsymbol{\epsilon}_t)$ is invertible.

The optimal value of the log-likelihood is

$$\begin{aligned} L(\hat{\boldsymbol{\alpha}}_n, \hat{\boldsymbol{\beta}}_n, \hat{\boldsymbol{\Sigma}}_n) &= -\frac{Mn}{2} \log(2\pi) - \frac{n}{2} \log |\hat{\boldsymbol{\Sigma}}_n| \\ &\quad - \frac{1}{2} \sum_{t=1}^n (\mathbf{z}_t - \hat{\boldsymbol{\alpha}}_n - \hat{\boldsymbol{\beta}}_n z_t^{(m)})' \hat{\boldsymbol{\Sigma}}_n^{-1} (\mathbf{z}_t - \hat{\boldsymbol{\alpha}}_n - \hat{\boldsymbol{\beta}}_n z_t^{(m)}), \end{aligned}$$

with

$$\hat{\boldsymbol{\Sigma}}_n = \frac{1}{n} \sum_{t=1}^n \{\mathbf{z}_t - \hat{\boldsymbol{\alpha}}_n - \hat{\boldsymbol{\beta}}_n z_t^{(m)}\} \{\mathbf{z}_t - \hat{\boldsymbol{\alpha}}_n - \hat{\boldsymbol{\beta}}_n z_t^{(m)}\}'.$$

Likelihood Ratio test

The sum in the log-likelihood writes

$$\begin{aligned}
 & \sum_{t=1}^n \text{Tr} \left\{ (\mathbf{z}_t - \hat{\boldsymbol{\alpha}}_n - \hat{\boldsymbol{\beta}}_n \mathbf{z}_t^{(m)})' \hat{\boldsymbol{\Sigma}}_n^{-1} (\mathbf{z}_t - \hat{\boldsymbol{\alpha}}_n - \hat{\boldsymbol{\beta}}_n \mathbf{z}_t^{(m)}) \right\} \\
 &= \sum_{t=1}^n \text{Tr} \left\{ \hat{\boldsymbol{\Sigma}}_n^{-1} (\mathbf{z}_t - \hat{\boldsymbol{\alpha}}_n - \hat{\boldsymbol{\beta}}_n \mathbf{z}_t^{(m)}) (\mathbf{z}_t - \hat{\boldsymbol{\alpha}}_n - \hat{\boldsymbol{\beta}}_n \mathbf{z}_t^{(m)})' \right\} \\
 &= \text{Tr} \left\{ \hat{\boldsymbol{\Sigma}}_n^{-1} \sum_{t=1}^n (\mathbf{z}_t - \hat{\boldsymbol{\alpha}}_n - \hat{\boldsymbol{\beta}}_n \mathbf{z}_t^{(m)}) (\mathbf{z}_t - \hat{\boldsymbol{\alpha}}_n - \hat{\boldsymbol{\beta}}_n \mathbf{z}_t^{(m)})' \right\}, \\
 &= \text{Tr} \left\{ \hat{\boldsymbol{\Sigma}}_n^{-1} n \hat{\boldsymbol{\Sigma}}_n \right\} = n \text{Tr}(\mathbf{I}_M) = nM.
 \end{aligned}$$

Hence, the optimal log-likelihood only depends on $\hat{\boldsymbol{\Sigma}}_n$ and writes

$$\mathcal{L}(\hat{\boldsymbol{\Sigma}}_n) = -\frac{Mn}{2} \log(2\pi) - \frac{n}{2} \log |\hat{\boldsymbol{\Sigma}}_n| - \frac{nM}{2}.$$

Likelihood of the constrained model

In the **constrained model** ($\alpha = 0$),

$$\mathbf{z}_t = \boldsymbol{\beta} z_t^{(m)} + \boldsymbol{\epsilon}_t, \quad t = 1, \dots, n$$

the GLS estimator of $\boldsymbol{\beta}$ is

$$\hat{\boldsymbol{\beta}}_n^c = \frac{\sum_{t=1}^n \mathbf{z}_t z_t^{(m)}}{\sum_{t=1}^n z_t^{(m)2}},$$

and $\boldsymbol{\Sigma}$ is estimated by

$$\hat{\boldsymbol{\Sigma}}_n^c = \frac{1}{n} \sum_{t=1}^n \{\mathbf{z}_t - \hat{\boldsymbol{\beta}}_n^c z_t^{(m)}\} \{\mathbf{z}_t - \hat{\boldsymbol{\beta}}_n^c z_t^{(m)}\}'.$$

The optimal value of the constrained log-likelihood is

$$\mathcal{L}(\hat{\boldsymbol{\Sigma}}_n^c) = -\frac{Mn}{2} \log(2\pi) - \frac{n}{2} \log |\hat{\boldsymbol{\Sigma}}_n^c| - \frac{nM}{2}.$$

LR statistic

$$\begin{aligned}\mathcal{L}(\hat{\Sigma}_n) &= -\frac{Mn}{2} \log(2\pi) - \frac{n}{2} \log|\hat{\Sigma}_n| - \frac{nM}{2}, \\ \mathcal{L}(\hat{\Sigma}_n^c) &= -\frac{Mn}{2} \log(2\pi) - \frac{n}{2} \log|\hat{\Sigma}_n^c| - \frac{nM}{2}.\end{aligned}$$

The **LR statistic** is

$$L_n = 2\{\mathcal{L}(\hat{\Sigma}_n) - \mathcal{L}(\hat{\Sigma}_n^c)\} = n\{\log|\hat{\Sigma}_n^c| - \log|\hat{\Sigma}_n|\}$$

and its asymptotic distribution under H_0 is the same as for W_n :

$$L_n \xrightarrow{d} \chi_M^2.$$

Critical region at the asymptotic level $\alpha \in (0, 1)$:

$$C = \{L_n > \chi_M^2(1 - \alpha)\}.$$

Data

- Monthly prices of $M = 6$ stocks
- Period of observation : August 2000 to October 2005 ($n = 63$).
- Riskfree asset : 3 months US Treasury bonds
- Market portfolio : the Dow Jones Industrial Average

These stocks represent 6 sectors of the Dow Jones Industrial Average index : American express (AXP), Citigroup (CITI, bank), Exxon (XOM, oil and gaz) , General Motors (GM, cars), Intel (INTEL, semi-conductors) and Pfizer (PFE, pharmaceutical sector).

Table – Performance of 6 stocks.

	AXP	CITI	XOM	GM	INTEL	PFE
$\alpha \times 10^3$	0.87	0.81	2.23	-2.41	-4.31	-5.21
p -value	0.72	0.76	0.40	0.59	0.52	0.06
β	1.23	1.20	0.52	1.44	2.28	0.46
$\beta^2 \sigma_m^2 \times 10^4$	5.77	5.48	1.04	7.91	19.8	0.80
$\sigma_\epsilon^2 \times 10^4$	3.50	4.18	4.22	12.0	26.7	4.64
Sharpe $\times 10^2$	-5.49	-5.36	5.05	-12.1	-13.2	-26.4

Wald test statistic value of the CAPM : $W_n = 1.20$. ($p\text{-val} > 0.975$)
 (to be compared with the quantiles of the $\chi^2(6)$)

Estimated Sharpe ratio for the DJ index : -10.6×10^{-2}

End of Chapter 3

Reminders on GLS estimation

In the regression model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U}, \quad E(\mathbf{U} | \mathbf{X}) = \mathbf{0}, \quad V(\mathbf{U} | \mathbf{X}) = \boldsymbol{\Omega} \text{ (positive definite)}$$

where $\mathbf{Y} \in \mathbb{R}^N$, $\mathbf{X}$ is a $N \times K$ matrix, the **Generalized Least-Squares (GLS)** estimator

$$\hat{\boldsymbol{\beta}}_{GLS} = (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{Y}$$

A **Quasi-GLS (QGLS)** estimator of $\boldsymbol{\beta}$ is

$$\hat{\boldsymbol{\beta}}_{QGLS} = (\mathbf{X}'\hat{\boldsymbol{\Omega}}_n^{-1}\mathbf{X})^{-1}\mathbf{X}'\hat{\boldsymbol{\Omega}}_n^{-1}\mathbf{Y}$$

where $\hat{\boldsymbol{\Omega}}_n$ is a consistent estimator of $\boldsymbol{\Omega}$.

► Return